

Project Design Phase

Solution Architecture

Date	15 February 2025
Team ID	Individual Project
Project Name	Book - A - Doctor
Maximum Marks	4 Marks

Solution Architecture

The **Book A Doctor** application follows a **Three-Tier Architecture** consisting of the Presentation Layer, Business Logic Layer, and Database Layer. The architecture is designed to provide a secure, scalable, and efficient healthcare appointment management system for patients and doctors.

The **Presentation Layer** is developed using **React.js** and provides an interactive and responsive user interface. Patients and doctors can access the platform through web browsers to perform various operations such as registration, login, appointment booking, appointment management, and medical report uploads.

The **Business Logic Layer** is implemented using **Node.js** and **Express.js**. This layer processes user requests, performs authentication and authorization, validates input data, manages appointment scheduling logic, and communicates with the database. JWT-based authentication ensures secure access to protected resources.

The **Database Layer** uses **MongoDB** for storing user information, doctor profiles, appointments, and medical reports. Mongoose is used as an Object Data Modeling (ODM) library to interact with MongoDB efficiently.

The system ensures secure communication between frontend and backend through RESTful APIs. Role-based access control is implemented to provide different functionalities for patients and doctors.

Components of the Architecture

1. Frontend Layer (React.js)

- Patient Dashboard
- Doctor Dashboard
- Login and Registration Pages
- Doctor Search Interface
- Appointment Booking Module
- Medical Report Upload Interface
- Appointment Tracking System

2. Backend Layer (Node.js & Express.js)

- Authentication Module
- User Management Module
- Doctor Management Module
- Appointment Management Module
- Medical Report Management Module
- REST API Services
- Role-Based Access Control

3. Database Layer (MongoDB)

- User Collection
- Doctor Collection
- Appointment Collection
- Medical Reports Collection

Data Flow

1. User accesses the Book A Doctor web application.
2. Patient or Doctor registers and logs into the system.
3. Authentication service validates credentials and generates JWT tokens.
4. Patients search for doctors and submit appointment requests.
5. Backend processes requests and stores appointment details in MongoDB.
6. Doctors view appointment requests and approve or reject them.
7. Appointment status is updated in the database and reflected on user dashboards.
8. Patients can upload medical reports which are securely stored and linked to their profiles.
9. All communication between frontend and backend occurs through secure REST APIs